

SSBMF Status Update - May 8, 2026

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1. Recent Improvements: Problems Addressed and Resolved

- **Risk direction is now corrected.** The 2026-05-05 Phase C sign correction fixed a major anti-concordance problem in both LB and YFB. This is now a resolved issue and the basis for interpreting all current external results.
- **Merged YFB collapse was resolved on 2026-05-06.** The 2026-05-05 report still showed merged YFB as effectively collapsed; the later 2026-05-06 update supersedes that result. Per-platform z-standardization plus removal of the rank transform recovered non-zero prognostic coefficients for merged TCGA+CPTAC.
- **Initialization constraints were tested and discarded.** Constrained singular value decomposition initialization and Cox-sign alignment did not improve ELBO or external prognosis.
- **Multi-start was tested and discarded.** A 30-restart sweep showed the standard singular value decomposition initialization already wins on ELBO, so routine restarts are not the current bottleneck.
- **K-selection is now a clearer methodological decision.** The 2026-05-06 work added an implemented K-fold CV pipeline and clarified what is currently reliable for LB versus still unresolved for YFB.

Brief note on YFB model development.

- **LB formulation:** prognosis enters through the latent loading matrix, $\eta = L\beta$.
- **YFB reformulation:** prognosis instead uses observed factor scores, $\eta = (YF)\beta$, which removes the LB train/test mismatch.
- **Derivation impact:** the survival term no longer acts through the original L -based predictor, so YFB required a separate Cox-on-YF derivation and YFB-specific update rules.

2. Current Results and Value Add: LB and YFB

Model	Training set	Prior(s)	Median external C by prior	Factor / beta summary
LB	TCGA-only	normal / point_normal	normal 0.610 / point_normal 0.610	LB priors agree; 2 active; max beta =0.0085
LB	CPTAC-only	normal / point_normal	normal 0.625 / point_normal 0.625	LB priors agree; 3 active; max beta =0.0152
LB	Merged TCGA+CPTAC	normal / point_normal	normal 0.644 / point_normal 0.644	LB priors agree; 3 active; max beta =0.0213
YFB	TCGA-only	normal / point_normal	normal 0.553 / point_normal 0.500	point_normal collapses; normal has near-zero betas
YFB	CPTAC-only	normal / point_normal	normal 0.561 / point_normal 0.500	point_normal collapses; normal has near-zero betas
YFB	Merged TCGA+CPTAC	normal / point_normal	normal 0.644 / point_normal 0.644	both priors agree; 3 active; max beta =0.0677

Notes: LB gives the same current external predictions under normal and point_normal. For YFB single-cohort fits, point_normal collapses to beta approximately zero, so the normal prior gives the only non-degenerate single-cohort result. Merged YFB uses the same TCGA+CPTAC training set, with per-platform standardization and no rank transform; under those settings, both priors agree.

- **LB is now the stable benchmark model.**
 - Median external C-index is **0.610** for TCGA-only, **0.625** for CPTAC-only, and **0.644** for merged training.
 - Best cohort result is **PACA-AU seq: 0.656-0.673**.
 - LB consistently identifies **2-3 active factors**.
- **Why only one prior is emphasized for LB.**
 - In the current LB benchmarks, **point-normal and normal give effectively identical external predictions**, so showing both is redundant; the document reports point-normal because it is the current default prior.
- **YFB remains mixed in single-cohort mode.**
 - With the normal prior, median external C-index is **0.553** for TCGA-only and **0.561** for CPTAC-only.
 - In the current single-cohort fits, the reported betas are effectively near-zero, so these models are not yet giving a convincing activated-factor solution even though some external C-indices are above 0.5.
- **Why only the normal prior is emphasized for single-cohort YFB.**
 - In the current YFB single-cohort benchmarks, the **point-normal prior collapses to beta 0**, so the normal prior is the only interpretable single-cohort YFB result. On the merged TCGA+CPTAC fit with the working preprocessing settings, **both priors agree**.
- **Merged YFB is currently the strongest “value add” result.**
 - On the same merged **TCGA+CPTAC** training set, using **per-platform standardization and no rank transform**, YFB achieves $K = 3$, $K_{\text{eff}} = 3$, $\max|\hat{\beta}| = 0.0677$, and external C-index **0.532-0.666** across all five cohorts.
 - Median external C-index is **0.644**, matching merged LB.

3. Open Questions and Decisions

K-selection path	Current status in repo	What recent results suggest
ARD from K_max	Partially present via shrinkage and K_eff behavior	Useful regularization, but not yet a sufficient standalone decision rule in PDAC
Bayesian cross-validation	Not yet implemented in a decision-ready form	Methodologically appealing, but currently a future development path
K-fold CV grid search	Implemented; currently the only operational selection workflow	Reliable for LB; still unreliable for YFB because of fold-level sign instability

- **Current LB recommendation.**
 - Under the 2026-05-06 K-CV sweep, the **normal prior selects $K = 3$** under the 1-SE rule.
 - The **point-normal prior selects $K = 8$** , but this is interpreted as a shrinkage/small- n artifact rather than strong evidence that 8 factors are biologically necessary.
 - The current practical compromise is $K = 5$ for LB.
- **Current YFB limitation.**
 - YFB K-CV is **not yet decision-ready**.
 - Point-normal collapses to **C = 0.5** in all folds; normal prior gives **anti-concordant C-index** across K because held-out folds do not yet have stable sign correction.
- **Main decisions for discussion.**
 - Should **K-fold CV remain the operational rule for LB** while YFB K-selection remains open?
 - Should the next methodological push be **Bayesian CV** or **ARD-style large- $K_{\max}$ shrinkage**?
- **Other open issues.**
 - **Single-cohort YFB still needs a scale-aware fix** analogous to the merged preprocessing fix that worked on the merged TCGA+CPTAC training set.
 - **LB still appears genomics-dominated in the loading update**, so survival influence on L may remain too weak even after sign correction.

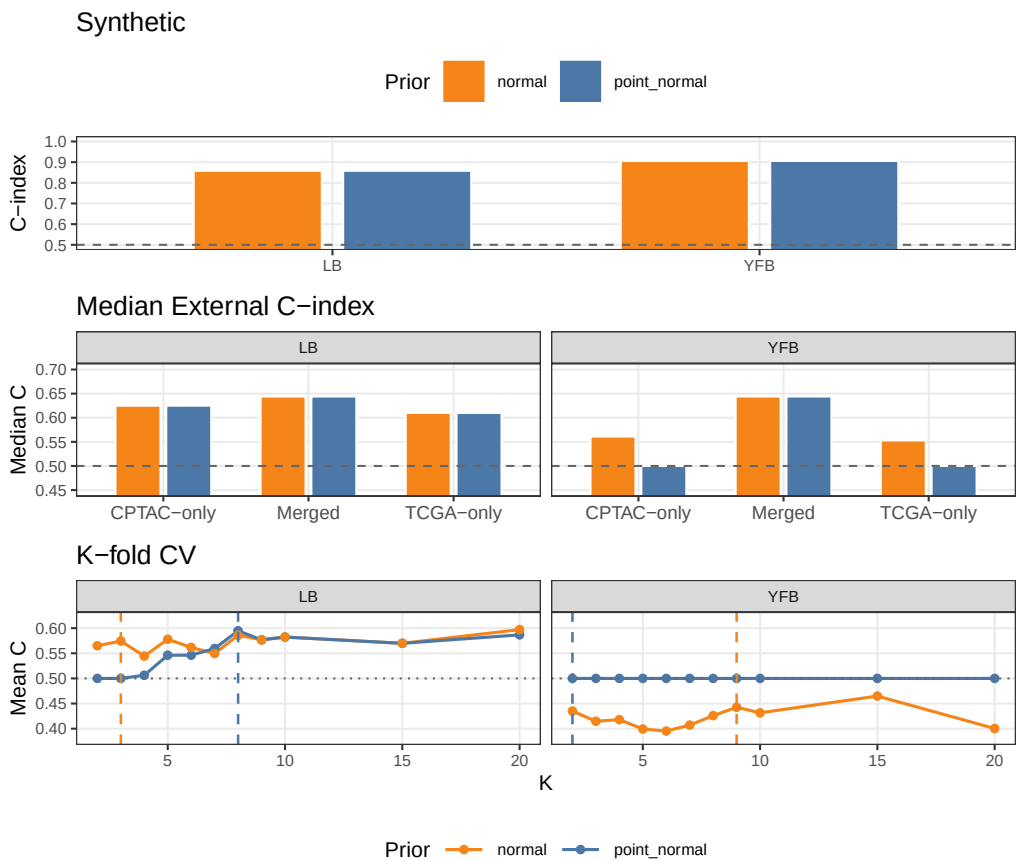


Figure 1: Current evidence across synthetic validation, PDAC external performance, and K-selection. Dashed lines mark selected K values from the 2026-05-06 sweep.